

GroupTrack

Quick Start, Installation & Troubleshooting

Tester and Developer Reference

Updated May 12, 2026

QUICK START

Before Your First Ride

1. Install GroupTrack on your Android device (see Installation section)
2. Power on your T1000-E Meshtastic radio
3. Open GroupTrack -- radio should connect via Bluetooth automatically
4. Verify connection: radio icon shows connected in the app

CRITICAL -- Two Radio Restrictions

RESTRICTION 1 -- T1000-E Firmware: DO NOT update beyond 2.6.11. The T1000-E Meshtastic radio firmware MUST stay at version 2.6.11 or earlier. Updating to any firmware beyond 2.6.11 will make the radio unstable for Bluetooth connectivity and radio configuration.

- Check current firmware: Meshtastic app > Radio Config > Device Info
- If firmware is 2.6.11 or earlier: DO NOT UPDATE
- If firmware was accidentally updated beyond 2.6.11: flash back to 2.6.11 using Meshtastic Flasher tool
- This restriction applies to ALL T1000-E radios used with GroupTrack

RESTRICTION 2 -- Keep Radio Within Bluetooth Range During Rides. ALWAYS keep your radio in your cart/vehicle within 50 feet of your phone. Do NOT carry the radio away from your phone beyond Bluetooth effective range.

- Bluetooth effective range: approximately 50 feet (15 meters)
- If radio exceeds BLE range while still "connected," the connection enters supervision timeout
- Symptoms: app freezes/pauses, position updates stop, UI becomes unresponsive
- The BLE connection does not cleanly disconnect -- it hangs in limbo waiting for the radio to reappear
- Recovery: force close GroupTrack, move radio back within range, reopen app
- Prevention: leave the radio mounted in your cart. Keep your phone in the cart. They travel together.

Pre-Ride Checklist

- ☐ Phone charged
- ☐ Radio charged and firmware at 2.6.11 or earlier
- ☐ Radio mounted in cart (stays with phone during ride)
- ☐ GroupTrack installed and opens without errors
- ☐ Radio connects via Bluetooth (solid connection icon)
- ☐ Map tiles downloaded for ride area (check blue overlay on Planning Map)
- ☐ Trail data loaded for ride area
- ☐

GPS lock acquired (position shows on map)

INSTALLATION

Installing GroupTrack

From Play Store (tester)

- Accept tester invitation link from ride organizer
- Download from Google Play Store
- Package: com.grouptrack.android

Play Store URL: <https://play.google.com/store/apps/details?id=com.grouptrack.android>

From Local APK (developer/tester)

```
# Android 1 -- V2.4 tester device
adb -s 8624SBCEDF00001789 install -r -d app-google-universal-debug.apk

# Android 2 -- V3.0 dev device
adb -s 24039703201775 install -r -d app-google-universal-debug.apk
```

INSTALL RULE: Always use -r -d flags. Never omit -r -d -- doing so wipes tile cache, app data, and config.

Play Store vs Local APK: Different signing keys. Must uninstall one before installing the other. Cannot have both installed simultaneously.

Post-Install Steps

1. Launch GroupTrack
2. Grant location permission when prompted ("Allow all the time" recommended)
3. Grant storage permission ("All Files Access")
4. Grant notification permission
5. If duplicate launcher icon appears: run `adb shell pm clear com.android.launcher3`
6. Open Planning Map and verify tile loading
7. Connect radio -- should auto-discover via Bluetooth

Radio Setup

1. Power on T1000-E radio
2. Verify firmware is 2.6.11 or earlier -- DO NOT UPDATE
3. Open GroupTrack -- radio should appear in device list
4. Select radio to pair
5. Verify connection: position icon and signal strength visible

Device Compatibility

The following devices have been tested with GroupTrack and T1000-E radios:

- Android 1 (8624SBCEDF00001789): VALIDATED - V2.4 tester device, stable BLE
- Android 2 (24039703201775): VALIDATED - V3.0 dev device (no GPS)
- P10_T (P10V07162601000362): VALIDATED - 3GB RAM, Android 16
- Tab8NEU / budget tablets: NOT COMPATIBLE - Cannot hold BLE to T1000-E. Supervision timeout 0x0008 after 8-10 seconds. Hardware issue, not app bug.

TROUBLESHOOTING

BLE Connection Issues

App freezes or pauses during ride

- Cause: Radio moved beyond Bluetooth range (~50ft) while still connected. BLE supervision timeout.
- Fix: Force close GroupTrack. Move radio back within range of phone. Reopen app.
- Prevention: Keep radio mounted in cart with phone. They must stay within 50ft at all times.

Radio connects then disconnects repeatedly (8-10 second cycle)

- Cause: Device hardware incompatibility (Tab8NEU, budget Android tablets). Supervision timeout reason 0x0008.
- Fix: Use a compatible device (see Device Compatibility table). This is a hardware issue, not an app bug.

Radio won't connect after firmware update

- Cause: T1000-E firmware updated beyond 2.6.11.
- Fix: Flash radio back to firmware 2.6.11 using Meshtastic Flasher tool. Do NOT use any firmware newer than 2.6.11.

Radio connects but position doesn't update

- Cause: Radio may be out of GPS fix, or BLE connection is degraded.
- Fix: Check radio has clear sky view for GPS. Force close and reconnect. Verify radio firmware is 2.6.11.

Map and Tile Issues

Map shows blank/grey tiles

- Cause: No tile cache for this area. Online tiles failed to load.
- Fix: Check connectivity. Download tiles for this area while on WiFi. Verify blue overlay on Planning Map.

Double labels on satellite map

- Cause: Old cached tiles with baked-in labels plus new overlay labels.
- Fix: Clear tile cache for affected area and re-download with current tile source URL.

Map tiles load online but show blank offline

- Cause: Tiles not downloaded for this area.
- Fix: Download tiles using Planning Map before going to the field. Verify blue overlay covers your ride area.

Trail Display Issues

No trails showing on map

- Cause: Trail data not loaded, or trails toggled off in Display Panel.
- Fix: Open Display Panel and toggle Trails ON. First toggle loads trail data from app assets.

App Issues

ANR (App Not Responding) on launch

- Cause: Blocking operation on main thread.

- Fix: Force close and reopen. If persistent, report with logcat output.

Duplicate launcher icons

- Fix: Run adb shell pm clear com.android.launcher3 to clear launcher cache.

App data lost after update

- Cause: Installed without -r -d flags, or switched between Play Store and local APK without uninstalling.
- Fix: Always install with adb install -r -d. Uninstall opposite signing key version before installing.
- Recovery: Spatial data in /sdcard/Documents/GroupTrack/data/ survives uninstall. Tile cache in /sdcard/Documents/GroupTrack/maps/tiles/ survives uninstall. Only app-internal data is lost.

Radio Firmware Reference

- Firmware 2.6.11: APPROVED -- Maximum recommended version for T1000-E
- Firmware 2.6.12+: DO NOT USE -- Destabilizes BLE and radio configuration
- Firmware < 2.6.11: OK -- Earlier versions work but may lack some features

Emergency Recovery

If everything is broken:

1. Uninstall GroupTrack
2. Flash T1000-E radio to firmware 2.6.11
3. Reinstall GroupTrack APK with adb install -r -d
4. Launch app -- spatial data in Documents/GroupTrack/ should still be there
5. Reconnect radio
6. Re-download tiles if cache was lost

DEVELOPER NOTES

Build Commands

```
# V2.4 debug build
./gradlew copyApkToDownloads

# V2.4 release APK
./gradlew assembleGoogleRelease -x uploadCrashlyticsMappingFileGoogleRelease

# V2.4 release AAB (Play Store)
./gradlew bundleGoogleRelease -x uploadCrashlyticsMappingFileGoogleRelease
```

Log Viewing

```
# Live convoy logs
adb -s 8624SBCEDF00001789 logcat | grep -E "ConvoyEngine|ConvoyApply|ConvoyMap"

# Pull debug log
adb -s 8624SBCEDF00001789 shell run-as com.geeksville.mesh.google.debug \
  cat files/convoy_debug.log
```

Asset Pull Rule

Use run-as cat redirect, never adb pull for internal storage:

```
adb -s [serial] shell run-as [pkg] cat files/[file] > app/src/main/assets/[file]
```

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T1000-E firmware: LOCK TO 2.6.11. Radio stays in cart during rides.